

Element A

Problem Identification

Individuals with hand arthritis often experience joint pain, stiffness, and reduced dexterity in their hands, making everyday tasks, such as gripping, writing, or opening containers, difficult and painful. Currently, there are no devices available to regular consumers beyond physical therapy and very expensive gloves. A solution is needed to combine features of heating, massaging, active grip assistance, and overall pain relief while staying affordable, designed for all hand arthritis sufferers.

Problem Definition

The glove must be designed for individuals with hand arthritis (such as osteoarthritis or rheumatoid arthritis), primarily affecting older adults or those with chronic hand inflammation. The glove should:

- Reduce pain and minimize stiffness during and after use
- Support finger mobility and enhance grip strength without restricting movement
- Be comfortable for prolonged daily wear (minimum 4–6 hours)
- Use soft, breathable, and non-irritating materials
- Be easy to put on and take off independently (even with limited hand strength or dexterity)
- Be affordable and potentially washable/reusable
- Optionally include therapeutic elements, like mild compression or heat retention

This issue of arthritis affects millions of people across the globe. According to the CDC, about a quarter of the US adult population (58 million people) has some form of arthritis, with the hands being one of the most affected areas. People older than 50 are the largest percentage of people with arthritis. Symptoms show up daily and often are worse in the morning or after long periods of not doing anything. Seasonal changes and cold weather also make symptoms worse. The people involved are typically older adults who have the conditions, caregivers, and healthcare providers, specifically rheumatologists and occupational therapists. Individuals with arthritis are struggling with managing pain and their symptoms by using medication as well as existing products such as compression gloves, heat packs, and other assistive devices. However, these existing gloves aren't meant for long-term daily use, aren't quality products, and aren't very comfortable. Arthritis is an issue globally, but is most prevalent in aging communities across North America, Europe as well and Asia. The struggle with arthritis is typically the most difficult at home; however, outside at work, in the community, the symptoms of arthritis remain a struggle. Arthritis pain is difficult constantly; however, it is the most difficult during cold weather, in the morning, after inactivity, and after using your hands for a long time. Arthritis causes inflammation in the

joints of the hand, leading to pain, stiffness, swelling, and loss of movement in the hand. Arthritis symptoms pop up daily; however, some experience the symptoms more than others, and some may experience worse episodes of pain and discomfort intermittently. In terms of unique situations, those elderly who live alone face the worst of the common difficulties as they have no one to help dress themselves, open jars, cook, write, etc. Working-age people with arthritis struggle with typing, writing, and anything where they need to use their hands significantly. Arthritis results in billions of dollars in healthcare costs globally, every year, and is a leading cause of loss of employment due to disability.

Problem Elaboration

Arthritis is the swelling of the joints. There are two main types—osteoarthritis and rheumatoid arthritis.

Osteoarthritis is the most common, affecting over 32.5 million US adults. It happens when the cartilage between the joints breaks down, leading to loss of flexibility, stiffness, and swelling.² Although osteoarthritis can damage any joint, the condition most commonly affects joints in the hands, knees, hips, and spine.³ People at risk include the old, obese, and those who experience repeated stress on the joints, like through sports. Females are more likely to get it, and the cause of this is unknown.

Rheumatoid arthritis is a chronic autoimmune disease that causes pain, swelling, and inflammation in the joints. But it can also damage other parts of the body, like the skin, eyes, lungs, heart, and blood vessels. It can increase the risk of heart disease.

Some people have both types; however, they do differ. Osteoarthritis causes damage to joints from overuse. Rheumatoid arthritis affects the lining of the joints and eats away at the bone under them. This causes a painful swelling that can cause joints to bend out of shape over time, called deformity.

If this issue of arthritis related disabilities persists, it could affect the stakeholder groups significantly. Patients will continue to exhibit pain and arthritis related symptoms severely inhibiting their daily life. Caregivers and family members will have to give up countless hours and much needed energy caring for these patients. Healthcare professionals and therapists will continue to deal with limited options in order to help individuals with arthritis live a painless life.

Overall, this issue exists in the larger context of a graying population within the world. Statistics show that globally, countries are facing higher dependency ratios, and a significantly higher median age. A graying population is associated with more arthritis patients. A graying population specifically in the US is forcing people to retire later as there are less social security benefits, with a smaller working population. A working population with hand arthritis faces productivity issues as many jobs require motor function which is limited with osteoarthritis. A

large population of people in pain while working becomes a societal problem that needs to be addressed.

In terms of mental health, people who feel like they are less self-sufficient due to higher difficulty in daily tasks from pain from osteoarthritis and rheumatoid arthritis, tend to have lower self esteem. They require assistance from caretakers and family members. Some people as they lose self-autonomy begin to feel more depressed as they feel like they are a dependency. This loss of independence is a major concern for those who are getting older and face these symptoms from arthritis. In terms of social participation, those suffering from arthritis tend to have less options when spending time with family and friends, and therefore begin to feel more isolated socially.

Concerns of Primary Stakeholder Groups

Primary Stakeholders:

1. Patients with Rheumatoid Arthritis (RA) and Osteoarthritis(OA)

- These are the target market, the consumers who rely on the glove for arthritis relief
- Will have the largest effect on sales
- Concerns
 - Due to safety concerns, intensive trials need to be conducted
 - Due to effectiveness concerns, significance test must be conducted, many trials
 - Due to comfortability concerns, comfortability of fabric, and weight of the product must take precedence
 - Due to pricing concerns, product should be as inexpensive as possible

2. Caregivers and Family Members (?)

- Could help patients use the glove, are a secondary market, as they may purchase the glove for patients
- Concerns
 - Same as patient concerns

3. Healthcare Professionals

- Doctors (especially rheumatologists), physical therapists, and occupational therapists.
- May recommend the glove as part of a treatment or rehabilitation plan.
- Their endorsement could drive clinical acceptance and insurance coverage.
- Also a secondary market
- Concerns
 - Same as patient concerns
 - due to competition concerns, product must do better than other products in the market

4. Investors / Founders

- Fund development and expect returns.
- Shareholders need to be paid for investments from revenue
- concerns
 - Same concerns as healthcare professionals
 - due to concerns of success, in depth market analysis must be conducted

Secondary Stakeholders (Individuals indirectly impacted or involved):

1. Health Insurance Providers

- Could cover the cost of the glove if medically approved.
- Have a stake in demonstrating reduced long-term costs from use.
- concerns
 - due to durability concerns, trials must be conducted to determine product life and longevity

2. Medical Device Regulators

- FDA (in the U.S.), CE (in Europe), and other certification bodies.
- Their approval is essential for legal distribution and claims.
- concerns
 - Due to concerns about product meeting regulations, regulations must be kept in mind when designing product

3. Retailers and Distributors

- Pharmacies, medical supply stores, and online platforms (like Amazon or DME suppliers).
- Will handle the sales channel and customer experience.
- concerns
 - due to distribution concerns, product must be easily shipped and distributed

4. Support Groups & Arthritis Foundations

- Nonprofits like the Arthritis Foundation may promote or validate the glove.
- They can help with outreach, community feedback, and partnerships.
- concerns
 - same as patient concerns

5. Researchers & Universities

- Could collaborate for clinical trials, new features, or grants.
- Help validate effectiveness and explore future applications (like AI-driven rehabilitation).(important: ethos marketing)
- concerns
 - effectiveness of product, studies and trials must be conducted

6. Manufacturers and Supply Chain Partners

- Produce, assemble, and deliver the physical components of the glove.(not really a stakeholder but is affected by the success of the product)

- concerns
 - due to production concerns product must be easily mass producible

Problem Justification

According to the CDC, approximately 53.2 million adults in the United States (21.2%) have diagnosed arthritis.¹ This number is expected to increase as the population ages² and the US shifts in terms of its demographics and population pyramid. Of these people, 1.5 million have rheumatoid arthritis³, and 32.5 million have osteoarthritis⁴—so out of all the people with arthritis, 64% have one of these types, exhibiting how important and prevalent the problem is in the US. Additionally, since the US does not offer free healthcare,⁵ most medications and devices are high-priced, and even if insurance covers the cost, the devices only offer a short-lived relief. With our device, patients will be able to access affordable, durable, and long-term relief from arthritis, especially in the hand, for everyday tasks.

Works Cited

1. "Arthritis." Centers for Disease Control and Prevention, 10 Jan. 2024, <https://www.cdc.gov/cdi/indicator-definitions/arthritis.html>.
2. "The Budget and Economic Outlook: 2024 to 2034." Congressional Budget Office, 7 Feb. 2024, <https://www.cbo.gov/publication/60875>.
3. "Rheumatoid Arthritis." Pfizer, <https://www.pfizer.com/disease-and-conditions/rheumatoid-arthritis>.
4. "Osteoarthritis Prevalence and Burden." University of North Carolina, OA Action Alliance, <https://oaaction.unc.edu/oa-module/oa-prevalence-and-burden/>.
5. "Mirror, Mirror 2024: Reflecting Poorly – Health Care in the U.S. Compared to Other High-Income Countries." The Commonwealth Fund, Sept. 2024, <https://www.commonwealthfund.org/publications/fund-reports/2024/sep/mirror-mirror-2024>.
6. "Arthritis and Rheumatoid Arthritis." Centers for Disease Control and Prevention, 26 Apr. 2024, <https://www.cdc.gov/arthritis/rheumatoid-arthritis/index.html>.
7. "Osteoarthritis." Mayo Clinic, 4 Oct. 2024, <https://www.mayoclinic.org/diseases-conditions/osteoarthritis/symptoms-causes/syc-20351925>.
8. "Osteoarthritis of the Hands." Arthritis Foundation, <https://www.arthritis.org/diseases/more-about/osteoarthritis-of-the-hands>.
9. "Rehabilitation of Older Adult Patients with Arthritis." Johns Hopkins Arthritis Center, <https://www.hopkinsarthritis.org/patient-corner/disease-management/rehabilitation-of-older-adult-patients-with-arthritis/>.
10. "PubMed – Arthritis Study." National Library of Medicine, <https://pubmed.ncbi.nlm.nih.gov/11005783/>.